Impulse Response and Poles and Zeros

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Key Concepts

1) Partial fraction expansion of the system function reveals the contribution of each pole to the impulse response.

    a) An pth-order pole at $z = 0$ contributes an impulse at time sample p.

    b) A single pole at d_k contributes an exponential term $A_k d_k^n u[n]$.

    c) A repeated pole at d_k contributes an exponential term multiplied by time $A_k n d_k^n u[n]$.

    d) A complex-conjugate pair of poles contributes an exponentially damped sine and cosine. The exponential damping

factor is given by the radius of the poles and the frequency is given by the angle.

2) The radius of the pole determines the time it takes for that term in the impulse response to decay.

    a) A pole at radius r contributes a term with envelope r^n.

    b) Poles close to the unit circle ($r = 1$) take much longer to decay than poles near the origin ($r = 0$).

3) In general, systems with poles near the origin have a faster response time than systems with poles near the unit circle.

Video Screenshots

0:08

How do the poles/zeros of an LTI system determine / impulse response? transient response?

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k] \quad \Longrightarrow \quad H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

$$h[n] \overset{z}{\longleftrightarrow} H(z)$$

Partial Fraction Expansion of $H(z)$

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{a_0 \prod_{k=1}^{N} (1 - d_k z^{-1})} = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1}^{N_1} \frac{A_k}{1 - d_k z^{-1}} + \sum_{k=N_1}^{N_2} \frac{C_k d_k z^{-1}}{(1 - d_k z^{-1})^2} + \cdots$$

$$h[n] = \sum_{r=0}^{M-N} B_r \delta[n-r] + \sum_{k=1}^{N_1} A_k (d_k)^n u[n] + \sum_{n=N_1}^{N_2} C_k n (d_k)^n u[n] + \cdots$$

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r^{th} order pole at zero $\longrightarrow$ $B_r \delta[n-r]$

Single pole at d_k $\longrightarrow$ $A_k (d_k)^n u[n]$

Double pole at d_k $\longrightarrow$ $C_k n (d_k)^n u[n]$

Example:

$$H(z) = 2z^{-2} + \frac{1}{1-\frac{3}{4}z^{-1}} + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)z^{-1}}{1+\frac{1}{2}z^{-1}}$$

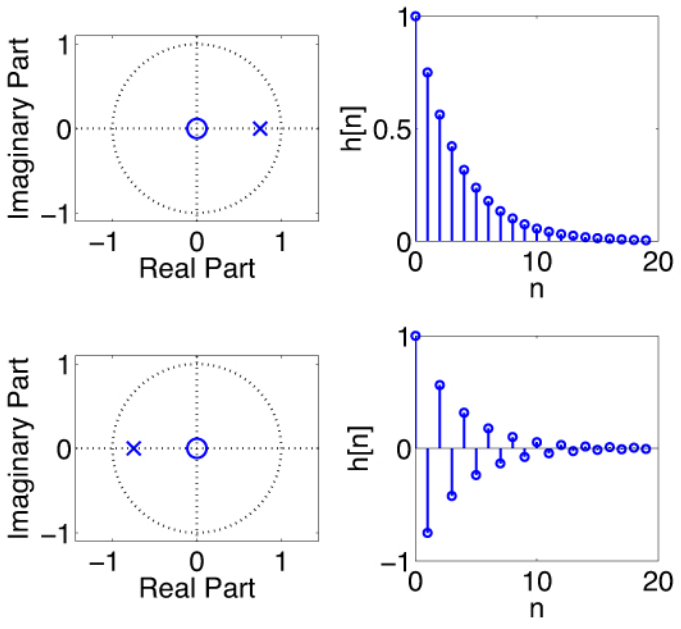
$$h[n] = 2\delta[n-2] + \left(\frac{3}{4}\right)^n u[n] + \frac{1}{2}n\left(-\frac{1}{2}\right)^n u[n]$$

Behavior of impulse response contribution depends on pole location and multiplicity

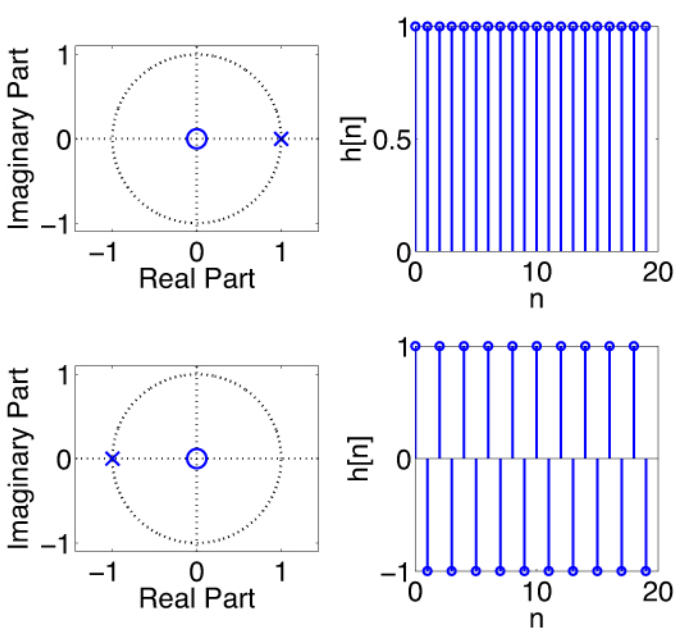
1) Single real pole

$$\frac{1}{1-az^{-1}} \longleftrightarrow a^n u[n]$$

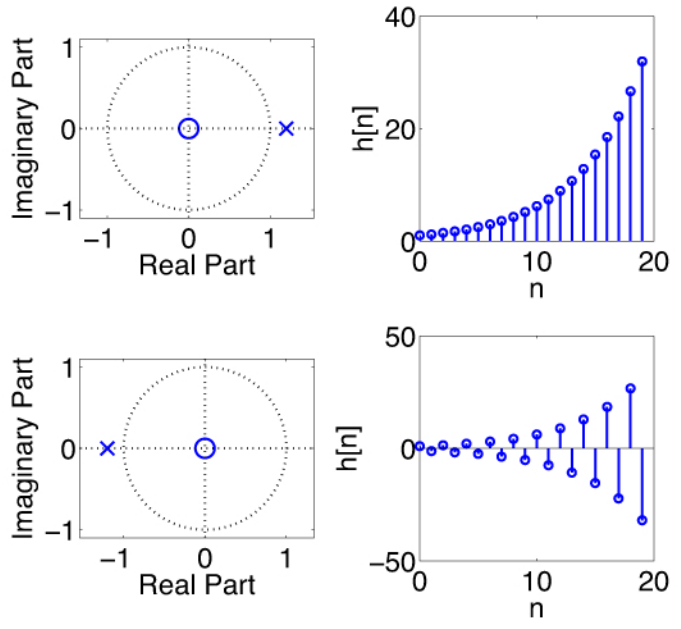
$|a| < 1$



$|a| = 1$



$|a| > 1$

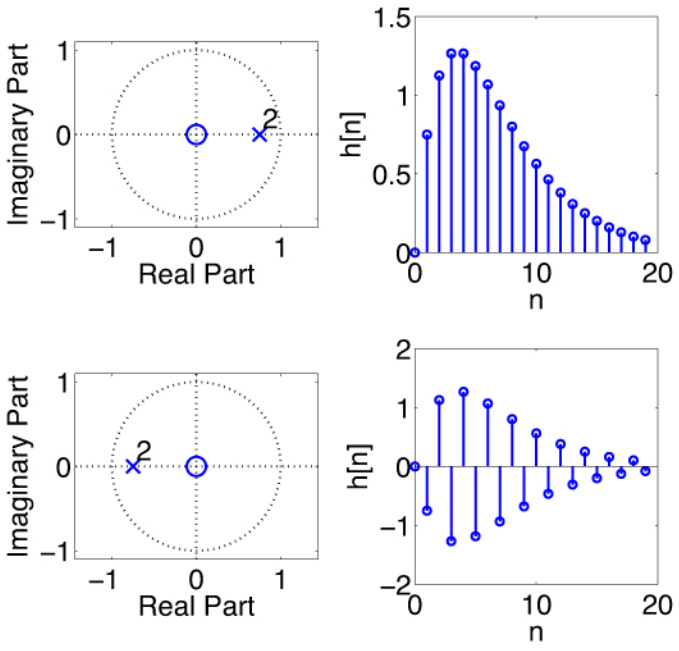


2) 2nd order real pole

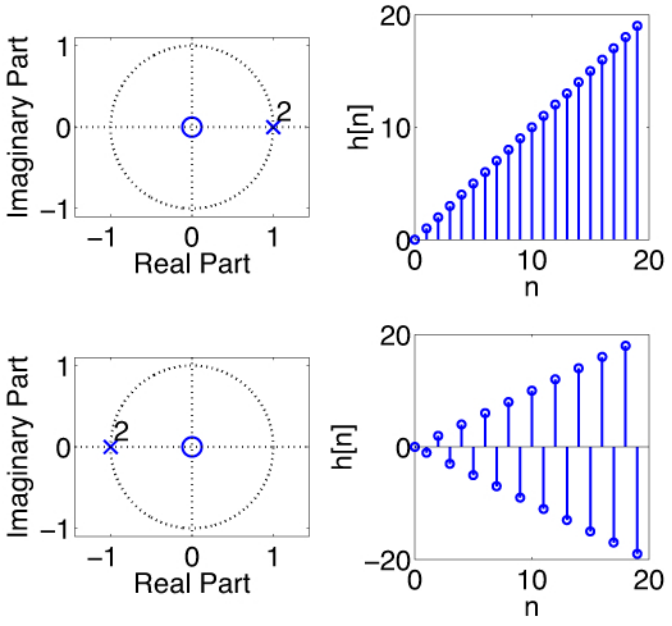
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$$\frac{a z^{-1}}{(1 - a z^{-1})^2} \longleftrightarrow n a^n u[n]$$

$|a| < 1$



$|a| = 1$



5) Complex conjugate pair of poles

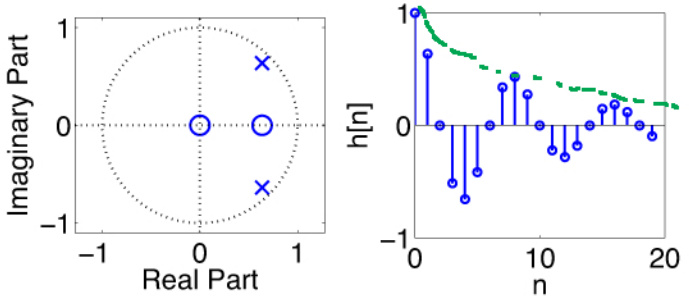
- poles at $r e^{\pm j\omega_0}$

$$\frac{1 - r\cos(\omega_0) z^{-1}}{1 - 2r\cos(\omega_0) z^{-1} + r^2 z^{-2}} \longleftrightarrow r^n \cos(\omega_0 n) u[n]$$

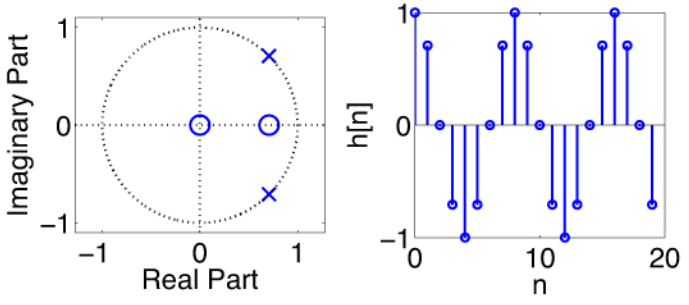
$r^n e^{j\omega_0 n}$
$r^n e^{-j\omega_0 n}$

$\omega_0 = \pi/4$

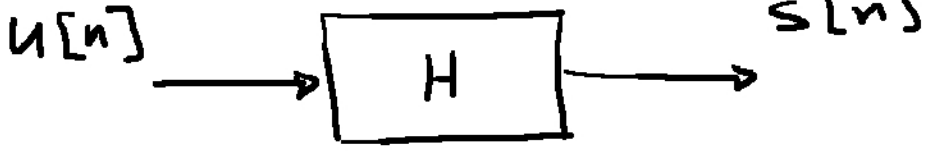
$|r| < 1$



$|r| = 1$



Impulse response duration → transient response time



"step response"

$$u[n] = \sum_{k=-\infty}^{n} \delta[k] \quad \Rightarrow \quad s[n] = \sum_{k=-\infty}^{n} h[k]$$

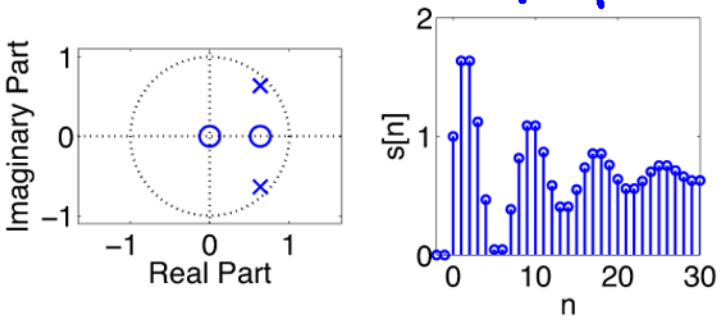
poles near

$z = 0 \Rightarrow$ fast response

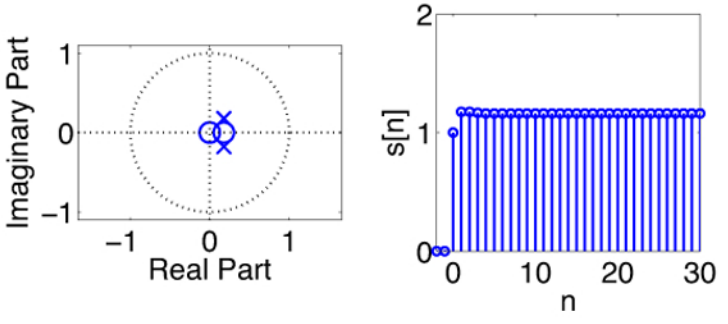
$|z| = 1 \Rightarrow$ slow response

step response

$r = 0.9$



$r = 0.25$



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